

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-aab63d8657c1ba587.jsonl`  
- SHA-256: `a5f3d3c8477d489ca10a616408a612e495cc8213d624c4de903062d11c624818`  
- Source modified: `2026-07-03T16:45:29+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:44:58.258Z)

You are a code-review VERIFIER. Repo: C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer, branch codex/alpha-readiness-plan. Sibling repo: C:/Users/avidu/Projects/khelsutra-guru/khelsutra.

CANDIDATE: docs/ALPHA_LAUNCH_READINESS.md §7 row A12 = "Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | Repo: khelsutra + badminton-highlight-indexer + collector | Depends: A1, A3 | Gated: Owner D13 reopen". Claim: this same flywheel work is ALREADY tracked in this repo as docs/COMPUTE_DECOUPLED_SERVING/README.md §7 row M11 ("Data-flywheel harness (D12): capture ? store ? shadow-eval ? promote to golden TRAINING set... skeleton landed: backend/flywheel.py... deps M7,M9, owner-gated consent"), and A12 does not mention M11 at all ? creating two trackers for one deliverable that will drift (e.g., M11's skeleton already landed while A12 says TODO).

Verify: (1) Read the full M11 row and §4.6 in docs/COMPUTE_DECOUPLED_SERVING/README.md. (2) Read A12 in context plus §2 D5 and §9 Q1 in docs/ALPHA_LAUNCH_READINESS.md. (3) Read khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md P10/P11 definitions. Decide: are M11 and A12/P10-P11 the SAME deliverable (engine-side capture/promote/re-eval flywheel) or genuinely different layers (M11 = engine harness; P10/P11 = khelsutra Studio UI surface for corrections)? Is the "will drift" cost real, or does A12's role as a decision row ("reopen or defer") rather than an implementation row make the overlap harmless? Would a one-line cross-reference to M11 in A12 resolve it?

Return exactly one verdict: CONFIRMED (same deliverable double-tracked, no cross-link ? quote both rows), PLAUSIBLE (partial overlap worth a cross-link), or REFUTED (different layers, no drift risk ? quote what distinguishes them). Final message: verdict word, 2-5 sentences of evidence with quotes, one-line suggested fix if CONFIRMED/PLAUSIBLE.

2. assistant (2026-07-03T16:45:00.189Z)

I'll verify this claim systematically by reading the relevant sections across both documents.

3. user (2026-07-03T16:45:00.635Z)

```
1      # Alpha Launch Readiness Plan  
2  
3      > **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**  
2026-07-03; **Last updated:** 2026-07-03  
4      > **Lifecycle:** `DRAFT` -> IN PROGRESS -> DONE -> archived`  
5      > **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed  
in `docs/README.md` and `docs/DOC_STATUS.md`.  
6      > **Relation to existing docs:** consolidates the launch-relevant pending items from  
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,  
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.  
7      > **Honesty note:** `[verified]` means checked against the local synced repos on  
2026-07-03. `[analysis]` means derived from a local command run on that snapshot. This is not
```

legal, financial, or launch approval.

8

9 ---

10

11 ## 0. TL;DR

12

13 The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. ``[verified]`` The golden manifest has 15 badminton videos; ``[analysis]`` the local manifest needed path rebasing before manifest-driven commands ran; ``[analysis]`` only 6 of 15 feature JSONs had the full fusion schema; ``[analysis]`` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.

14

15 The first controlled alpha should be a hosted, authenticated product loop: upload a video, process it, select rallies, compile, and download a reel. The owned-model/commercial rally-detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB checkpoint provenance is resolved.

16

17 ## 1. Problem & goal

18

19 The previous launch blockers were spread across roadmap docs and mixed together three different questions:

20

21 - ****Can an invited tester use the product without SSH?****

22 - ****Can the 15-video golden corpus now support honest quality gates?****

23 - ****Can we claim or ship an owned/commercial rally detector?****

24

25 This doc separates those into executable workstreams. The goal is to reach an invite-only alpha where the product is usable end-to-end, the quality claims are honest and reproducible, and the remaining model/legal gates are explicit rather than accidentally implied by launch activity.

26

27 ## 2. Decisions locked

28

29 | # | Decision | Source / date | Implication |

30 |---|---|---|---|

31 | D1 | ****Product alpha and owned-model launch are separate decisions.**** |

``NEXT_STEPS.md`` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without claiming the owned model is commercially cleared. |

32 | D2 | ****First alpha must be page-driven, not SSH/CLI-driven.**** | ``NEXT_STEPS.md`` P0 UI-driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling, compile, and download are launch-critical. |

33 | D3 | ****The 15-video corpus is now a measurement asset, not a free launch pass.**** | ``data_pipeline/GOLDEN_VIDEOS.md``; local eval runs, 2026-07-03 | Rebase/portability, feature completeness, baseline refresh, and gate ratification come before quality claims. |

34 | D4 | ****Do not promote served Gate A by default from the current n=15 evidence.**** | ``backend.eval.serve_contrast`` run, 2026-07-03 | Proposal-side lift does not survive served-windowing safety; keep it as an opt-in/analysis lever. |

35 | D5 | ****P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**** | sibling ``khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md``; verified 2026-07-03 | First alpha can use manual/offline corpus promotion; automated flywheel follows after serving and gates are stable. |

36

37 ## 3. Verified snapshot, 2026-07-03

38

39 All five Git checkouts were fast-forwarded before analysis:

40

41 | Repo | Branch | Synced HEAD |

42 |---|---|---|

43 | ``badminton-highlight-indexer`` | ``master`` | ``e4914d1`` |

44 | ``khelsutra`` | ``master`` | ``b0f70a5`` |

45 | ``rally-annotator`` | ``main`` | ``ef9e088`` |

46 | ``sports-data-collector`` | ``master`` | ``bfc014f`` |

47 | ``sports-obsreport`` | ``main`` | ``a0c7252`` |

48

```

49     Corpus/eval facts:
50
51     - `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says
`output/human_lovo_manifest.json` is the ingested golden corpus at `n = 15`.
52     - `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest
found all 15 trajectories and labels in this workspace.
53     - `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54     - `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only
6 videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
55     - `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
56     - `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
57     - `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15>
--features-dir output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off
tolF1 `0.1382`, but marked the frozen baseline stale due to config/corpus changes.
58     - `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity
group` currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
59
60     ## 4. Launch strategy
61
62     ### 4.1 Alpha lane: make the product usable
63
64     The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
65
66     ### 4.2 Quality lane: make the 15-video corpus gate-ready
67
68     The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
69
70     ### 4.3 Flywheel lane: make tester feedback useful
71
72     The flywheel lane is successful when user corrections can flow into the collector/vault
path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning,
but it should not block the first controlled alpha if manual promotion can bridge the gap.
73
74     ## 5. Risks and guardrails
75
76     | Risk | Why it matters | Guardrail |
77     |---|---|---|
78     | Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
79     | Stale local manifests create false confidence | Manifest-driven tools skipped or
failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke
check |
80     | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still
n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
81     | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly
reported stale baselines | Refresh baselines only after reviewer sign-off and record the
command/output |
82     | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but
can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha
lane is already stable |
83     | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained
model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product
launch tasks |

```

```

84
85     ## 6. Honest limits
86
87     Completing this doc's alpha lane does **not** prove the owned model is ready, that WASB
weights are commercially clean, that the product can serve unbounded public traffic, that multi-
sport is ready, or that every tester correction automatically improves the model.
88
89     Completing the quality lane does **not** by itself deploy a service. It only makes the
measurement surface honest enough for a launch decision.
90
91     ## 7. Deliverables & progress tracker
92
93     Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
94
95     | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96     |----|-----|-----|-----|-----|-----|----|
97     | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |
98     | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelesutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99     | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |
100    | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101    | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
102    | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103    | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104    | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105    | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106    | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
107    | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
108    | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109    | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelesutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110    | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |
111
112     ## 8. Concrete next PR sequence
113
114     1. **Docs PR:** land this tracker.
115     2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
116     3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.

```

```

117     4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118     5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119     6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
120
121     ## 9. Open questions
122
123     1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124     2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125     3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
126     4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
127     5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
128
129     ## 10. Definition of done
130
131     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
132     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
133     - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
134     - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
135     - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
136     - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
137     - [ ] Ship-gate target and allowed alpha claims are recorded.
138     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
139     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
140     - [ ] Owner records GO/NO-GO for the first alpha wave.
141
142     ## 11. References
143
144     Internal docs:
145
146     - `docs/NEXT_STEPS.md`
147     - `docs/ROADMAP.md`
148     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
149     - `docs/R2_EVAL_METRIC_DESIGN.md`
150     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
151     - `docs/COMPUTE_DECOUPLED_SERVING/README.md`
152     - sibling `khehsutra/NEXT_STEPS.md`
153     - sibling `khehsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
154     - sibling `khehsutra/docs/PER_RALLY_SELECTION.md`
155     - sibling `sports-data-collector/data/golden_source_videos.json`
156
157     Local verification commands used for the 2026-07-03 snapshot:
158
159     ```powershell
160     python -m backend.eval.fusion_compare --features-dir output
161     python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
162     python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor
eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
163     python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir
output --no-report
164     python -m backend.eval.ablation --features-dir output --granularity group
165     ```
166

```

```
167 ---
168
169 ### Changelog
170
171 - 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`,
golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
172
```

4. user (2026-07-03T16:45:00.775Z)

```
1      # Compute-Decoupled Rally Serving ? one pure core, profile-selected compute + storage
2
3      > **Status:** ? **M0 APPROVED** (owner, 2026-06-21) ? **IN PROGRESS (build M1+)** .
**Owner:** Avi . **Created:** 2026-06-21 . **Last updated:** 2026-06-21
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to
`docs/archives/past_projects/` when DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory
`current-state` + `docs/NEXT_STEPS.md` once work starts.
6      > **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
7      > **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed
**compute-decoupled serving** because **truly-local is a co-equal, first-class profile**, not an
afterthought.
8      > **Honesty note:** claims tagged `[verified]` (checked against code by the design
workflow `wf_56274ff0`), `[design]` (proposed), `[researched]`.
9
10     ---
11
12     > **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video
? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next
to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user
*confirms* (D15).
13
14     ## 0. TL;DR
15     The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three
deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch
on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process
vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB
/ mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local**
(local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive /
bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus
**cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the
same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the
**parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.
16
17     ## 1. Problem & goal
18     - **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4)
and is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero +
cost guard)** become this project. The owner needs **both** a hosted service *and* a truly-local
mode, with the **core (detect+stitch) unaffected** by which one runs.
19     - **The two user-facing outcomes** (owner, 2026-06-21):
20     - ***(a) Hosted service** ? operate on small **local uploads** (`<2GB`) **and* load from
**gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching**
(stitching is compute-intensive ? run it where its cost is accounted for).
21     - ***(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local
GPU**, run inference **and** stitching **completely locally**.
22     - **"Good" looks like:** the same `video ? rallies ? reel` core gives identical windows
+ stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup
choice**, never a code branch in the core.
23
24     ## 2. Decisions locked
```

25

26 | # | Decision | Source | Implication |
27 |---|-----|-----|-----|
28 | D1 | The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile**
branch inside it. | design `[verified]` | All profile difference lives in config + the two
registries. |
29 | D2 | **Two registries**: `ComputeBackend`` (where) selected by `deployment.profile`` +
`compute_target``; `StorageBackend`` (what-bytes) by `input_backend`/storage.backend``. | ADR-014
| Realizes PLATFORM_ARCHITECTURE's `local\|hosted\|byo_worker``. |
30 | D3 | **Profiles shipped**: `truly-local``, `cloud-serving``, `cpu-mock`` (CI);
`byo_worker`` reserved/deferred. | ADR-014 | Enum/seam reserved so BYO isn't precluded. |
31 | D4 | In **cloud-serving**, STITCH runs on Cloud Run (co-located with detect) so
CPU/wall + egress are metered into the per-job cost ledger. | owner (a) | Stitch-on-laptop would
hide real cost. |
32 | D5 | **Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory);
detect is byte-exact only same-GPU. | deploy report 2026-06-20 | Cross-GPU asserted at the
rally/window level, never CSV bytes. |
33 | D6 | **Default-OFF**, smallest-safe-first, one PR per milestone; any core-caller
change (e.g. configurable output base) ships behind a flag + a Launch Report. | `[launch-report-`
`convention]` | Zero-regression discipline. |
34 | D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an
async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that
~\$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane
only on a UX complaint. |
35 | D8 | **Pricebook** = a versioned DB table; cost computed in **USD** (matches GCP
billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore
market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |
36 | D9 | **Edge auth** = Cloudflare Access (reuse the site's existing CF). Lock the Cloud
Run **ingress** to the CF proxy + **verify** the `Cf-Access`` JWT signature (so a direct hit to
the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity
system. |
37 | D10 | **Free-tier "data-for-reels" trade**: free reels in exchange for **training**
rights (uploaded footage + derived trajectories/labels); **paid tier** = no-train (privacy is
the paid feature). **Carve-outs**: train ONLY on attested **ADULT** footage (minor footage ?
reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on
DERIVED data + **auto-delete raw**; **ToS counsel-reviewed**; live training-on-uploads
gated to M9 (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties
`[paper-coauthor-aim]` / PERSONALIZATION flywheel. |
38 | D11 | **Quality floor: Gemini handoff ON** for cloud-serving until the owned model
clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the
owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a
training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |
39 | D12 | **Data-flywheel harness** is IN SCOPE (owner add, Q5): capture the (consented,
adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video
workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones**
to the golden TRAINING set (human-reviewed labels) so the owned model climbs toward the D11
floor (then Gemini flips off). | owner 2026-06-21 | new **M11**; ties D10 (consent) +
`data_pipeline/`` + the eval/experiment harness. |
40 | D13 | **Profile is user-switchable ANYTIME** ? a headline FEATURE. The same user runs
local today, cloud tomorrow, back to local ? it's just a config/startup choice; the pure
core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ?
your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the
rally/window level, not bit-identical across GPUs (D5); a single job runs in one profile,
switching is **between** jobs. |
41 | D14 | **NORTHSTAR** ? operate toward this: a **mobile, app-like** flow ? point at a
Google Drive video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel
lands **back in Drive**, next to the source. | owner 2026-06-21 | Implies a `gdrive-api``
(OAuth) StorageBackend (read source + write reel back) ? distinct from the local **mount**
backend; **resolves \$8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must
support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive
write); not necessarily v1, but the design must not preclude it (the `StorageBackend`` ABC
accommodates it). |
42 | D15 | **Auto-detect SUGGESTS**, the user **CONFIRMS** ? execution is NEVER a surprise. A
GPU owner may still choose cloud; **cloud (= spend)** is never entered silently; the active
profile is always explicit + surfaced. | owner 2026-06-21 | Amends \$4.3 (auto-detect = a

proposed default, not an auto-decision). |

43 | D16 | ****M6/M7 implementation SHAPE + the cheap \$8 picks (owner 2026-06-21):**** (a) ****Cloud Run JOBS**** (not Services-with-GPU) for the detect+stitch unit, ****bucket-poll** for completion** (reuses ``JobWaiting`/`claim_due_job`` verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) ****stitch ENCODE = libx264**** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? ****NVDEC** for DECODE only** (a speed lever that never touches output bytes); ****NVENC** encode is DEFERRED behind a config knob**, revisit only if M7 cost data justifies relaxing parity (\$8 Q8); (c) ****input cap = hard-reject ``<2GB``**** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? \$8 Q9). | owner 2026-06-21 | M6/M7. Resolves \$8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |

44 | D17 | ****Real cloud verification BUDGET = \$100 / ?10k cap (owner 2026-06-21):**** the owner authorized actual GCP spend for due-diligence + production-grade testing/verification of M5/M7 (real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), ****provided the cap is respected**** ? confirm exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate GPU, and ****DELETE every resource after**** (`[[gcp-vm-always-delete]]`). Removes the M6/M7 "spend" gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12) stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code) within the cap. |

45

46 **## 3. Foundation ? evolution / ADR lineage ? *trace the idea end-to-end***

47 This project is the latest step in a long decision chain; each ADR contributed a piece and a ****subtlety that carries forward****:

48

49 | ADR / doc | Contributed | Subtlety carried into this project |

50 |---|---|---|

51 | ****ADR-005**** | Permissive licensing (MIT/Apache-2.0/BSD only) | The engine + any served model + the ****container image**** (ffmpeg, fonts, weights) stay commercial-clean. |

52 | ****ADR-008**** | Owned-model topology; ****train==serve** featurizer single-sourcing** | The "core unaffected across profiles" is the same parity discipline, extended from train/serve to ****local/cloud****. |

53 | ****ADR-009**** | KhelSutra Guru; ****local-first delivery**** | The ****truly-local profile**** is the direct continuation ? privacy/offline self-host stays first-class. |

54 | ****ADR-010**** | DECOUPLED: D0?GCP pivot | The cloud-compute origin (GPU + co-located bucket). |

55 | ****ADR-011**** | DECOUPLED scope = M0?M2+M3.5; ****deferred M3 (serving) + M4 (billing/cost-guard)****; overrode the "rent nothing / not a service" posture | ****This project realizes M3/M4.**** Carries \$06's owner-gated gates: ****DPA/consent for non-owner uploads, collector `md5` keying, WASB-C3****. |

56 | ****ADR-012**** | GPU-native; TPU deferred; ****Cloud Run+GPU scale-to-zero = serving model****; ****NVDEC**** named as the decode lever | The serving substrate + the 7-step seed scope this expands. |

57 | ****ADR-013**** | Drop D0; ****GCP sole compute stack****; \$200 D0 credits ? non-compute only | The provider for cloud-serving; D0 Spaces remains only a possible S3 *storage* target. |

58 | ****? ADR-014 (new)**** | ****Compute-decoupled serving:**** one pure core, profile-selected compute+storage; stitch-where-metered | This doc. ****Refines**** ADR-011/012, does ****not**** supersede them. |

59 | PLATFORM_ARCHITECTURE.md | The ``ComputeBackend`/`StorageBackend`` registries | The abstraction this realizes in code. |

60 | VIDEO_LOCALITY_MODEL.md / DESKTOP_APP_PLAN.md | L0 fully-local tier / ``LocalDesktop`` backend | The truly-local profile = these, productionized. |

61 | 07-COMPUTE-ABSTRACTION (archived) | ``DeviceContext`/`DeviceProfile`` (designed) | WASB hardcodes ``device='cuda'`` with ****no CPU fallback**** ? a portability gap M4 closes. |

62 | DEPLOY_REPORT_GCP_2026-06-20 (archived) | First real L4 run; ****cross-GPU sub-pixel parity finding**** | Why D5 (parity at window level, not CSV bytes). |

63

64 **## 4. Design / architecture**

65 See [``architecture.svg``](architecture.svg) for the picture. ****The core never changes; the box around it does.****

66

67 **### 4.1 The core contract (must stay byte/behaviour-identical) ``[verified]``**

68 1. ****DETECT (GPU):**** ``DetectorRunner.run_predict(video_path, output_dir, video_id)`` ? CSV[Frame,Visibility,X,Y]` (``backend/pipeline/detectors/base.py:164``). Three interchangeable impls ? ``WasbRunner`` (WSL), ``NativeWasbRunner`` (Linux GPU), ``StubDetectorRunner`` (CPU mock) ? emit the ****identical CSV schema****. The trajectory CSV is the stable artifact boundary.

69 2. ****WINDOW (CPU pure fn):**** ``trajectory_to_action_windows(points, fps, frame_width, ?)``


```

? windows` (`tracknet_runner.py:281`). A shared `@staticmethod`, zero GPU/IO/randomness ? same
trajectory + config = byte-identical candidate windows. Used by every runner *and* the
eval/regression stack verbatim.
70     3. **STITCH (CPU/ffmpeg):** `VideoCompiler.compile_highlights(raw_video, segments,
out_name)` ? reel` (`stitcher/compiler.py:303`). Pure FS+subprocess (merge ? ffmpeg ? watermark
? per-clip libx264 trim ? concat). Deterministic for a given input + ranges + watermark config.
71
72     **Non-goal (the trap to avoid):** *no* code branch inside detect/window/stitch keyed on
profile ? the same discipline the experiment harness already enforces (a disabled cue is byte-
identical to CONTROL).
73
74     ### 4.2 The deployment profiles
75     | Profile | Compute | Storage | Orchestration | When |
76     |---|---|---|---|---|
77     | **truly-local** | `compute_target=local_gpu`; `detector_impl=native` (Linux) / `auto`
(WSL); stitch **in-process** | `local` (or `gdrive` = mounted Drive) | `python -m backend.main`
or `worker infer` CLI; existing async job worker | Self-host / offline / privacy / dev-on-a-GPU-
box ? owner (b) |
78     | **cloud-serving** | `compute_target=cloud_run`; `detector_impl=native` (L4, cuda);
**detect AND stitch on the same Cloud Run job** | `gcs` (or gdrive mount / assembled upload);
per-video workspace; reels via signed URL | Cloud Run control plane enqueues ? Cloud Run GPU job
pulls/runs/pushes; `WAIT_AND_RESUME` reschedules the control-plane job; scale-to-zero | Hosted
SaaS ? owner (a) |
79     | **cpu-mock** | `compute_target=cpu_mock`; `detector_impl=stub` (synth/replay) |
`local`/in-memory fakes | `run_all_tests.py` / pytest | CI, regression, profile-parity, GPU-free
dev |
80     | *byo_worker* (deferred) | tenant GPU via outbound pull agent | tenant's own bucket
(signed URLs) | long-poll pull; same job table scoped by tenant | Privacy-sensitive enterprise ?
**DEFERRED**, enum/seam reserved |
81
82     ### 4.3 Profile selection (the "startup/setup config") `[design]`
83     ONE new top-level `deployment` section on `AppConfig` + preset application + env auto-
detect in `load_config`:
84     - `deployment.profile` ? {truly-local | cloud-serving | cpu-mock}` (empty = today's
manual knobs, **fully backward-compatible**).
85     - Selecting a profile applies a **coherent preset bundle** of *existing* knobs that
today drift independently: **INPUT** (new `input_backend`/`input_root`, parallel to the output-
side `StorageConfig`), **COMPUTE** (new `compute_target` enum locking `detector_impl` +
`skip_ai_handoff` + workspace strategy), **STORAGE**
(`storage.backend`/`workspace_root`/`single_folder_per_video`, all already exist).
86     - **Precedence:** built-in defaults ? profile preset ? `config.json` ? env
(`RALLY_DETECTOR_IMPL`, `WASB_*`, `DEPLOYMENT_PROFILE`, ?) ? worker `--set k=v`. Every knob
stays individually overridable.
87     - **Env auto-detect SUGGESTS, the user CONFIRMS (D15 ? no surprise execution):** the env
gives a *proposed* default (`K_SERVICE` ? cloud-serving; `CI=true` ? cpu-mock; else
`torch.cuda.is_available()` ? truly-local), but the user **explicitly confirms or overrides** it
? a GPU owner may still pick cloud, and **cloud (= spend) is NEVER entered silently**. The
active profile is always surfaced (logged/shown). **Per-profile startup checks fail/warn fast**
(cloud-serving without GCS creds/GPU ? loud error before any job).
88     - The worker (`cmd_infer`/`cmd_train`) is **already profile-agnostic** (reads
`config.storage`, accepts `--config`/`--set`) ? no worker rewrite.
89
90     ### 4.4 Compute placement ? and *why stitch runs on Cloud Run* `[design]`
91     DETECT on GPU everywhere; WINDOW CPU-pure everywhere; STITCH is real CPU work. In
**cloud-serving, stitch is co-located in the Cloud Run job** so `gpu_active_seconds` (detect) +
`wall_seconds` (detect+stitch) + `bytes_out` (reel egress) land on **one metered invocation** ?
the per-job cost ledger. Running stitch on a laptop would leave that cost
**invisible/unattributed** ? exactly what the owner wants to avoid. (Stitch stays synchronous in
the GPU job for the common one-reel case; a queued CPU-only stitch fallback only if compiles get
bursty ? the job table + `claim_due_job` already supports both.)
92
93     ### 4.5 Reuse map ? most of the seams already exist `[verified]`
94     **Exists:** the 3-stage core; `_resolve_runner` (the single compute seam);
`StubDetectorRunner` (`replay_csv` = byte-exact $0 anchor); `StorageBackend` ABC + 4 backends +
`StorageRef.parse_uri`; the worker CLI (fetch/put, `--config`/`--set`); the async job control

```

plane (`jobs` table, `claim\_due\_job` CAS); `ExecutionPolicy WAIT\_AND\_RESUME`/`JobWaiting` (carries to Cloud Run dispatch verbatim); the chunked-upload router (`<2GB`); `skip\_ai\_handoff` (LocalNoAI); storage fakes; golden fixtures + experiment harness. ****Partial:**** per-video workspace helper (default-OFF); `DeviceContext` (designed); edge-auth (v4 `user\_id` columns exist). ****Net-new:**** the `deployment` section; `input\_backend`; configurable output base (kill hardcoded `output/`); Cloud Run dispatch shim + container image; cost ledger + pricebook; edge-auth header extraction; startup env detection.

95

96 ### 4.6 The data-flywheel harness (D12 / M11) `[design]`

97 Gemini-on (D11) ships good reels now; this harness is ****how we earn the right to flip Gemini off****. On a ****consented, adult**** cloud-serving job (the D10 trade), the pipeline additionally:

98 1. ****Captures**** the source upload + the Gemini reel + the rally output (intervals/report) into the per-video workspace (`workspaces/<video\_id>/`), instead of the no-consent auto-delete path.

99 2. ****Reliably stores**** them durably in the bucket (the consented-retention bucket; raw video kept only for the training pool, not the no-consent path).

100 3. ****Runs shadow evals**** ? the owned model scores the same video alongside Gemini (`backend/eval/experiment.compare` + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").

101 4. ****Promotes to golden**** ? captured videos that pass review become ****golden TRAINING rows**** (human-reviewed labels via the annotation flow ? `data\_pipeline/`), growing the corpus ? the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.

102 This is the consent?capture?eval?goldenize ****flywheel**** (the cloud realization of the PERSONALIZATION contribution model): every consented free reel makes the owned model better. Reuses `data\_pipeline/`, `backend/eval/`, and the M9 consent gate; ****adults-only****, raw-retention gated by the D10 consent.

103

104 ## 6. Honest limits ? what this does NOT do / prove

105 - A green CI suite proves ****plumbing + determinism (window+stitch) + profile-parity**** for \$0 ? it does ****NOT**** prove field quality, nor that real-GPU detection is good (that's owner-side, real-video).

106 - ****Cross-GPU detection is NOT byte-identical**** ? parity is a window-level claim, not a CSV-byte claim.

107 - The cloud `0.1.0-l4` weights are ****n=2 plumbing-grade**** ? cloud-serving ships with ****Gemini handoff ON**** until the owned model clears a floor (open Q).

108 - This does not resolve ****WASB-C3**** (sharing a trained model) or the ****DPA/consent**** gate ? both remain owner/counsel-gated before public, non-owner serving.

109

110 ## 7. Deliverables & progress tracker ? ****source of truth****

111 Legend: ? Todo · ? In progress · ? Done · ? Gated. ****One small PR per row****, off `origin/master`, no stacking. Default-OFF where it touches core callers.

112

ID	Deliverable	Depends	Gated?	Status	PR
M0	This design doc + project dir + `runlogs/` + ADR-014			?	owner sign-off ?
? APPROVED 2026-06-21					
M1	`deployment.profile` + `compute_target` config + preset bundles + precedence/auto-detect; unit tests (no behaviour change, profile='' default)			M0 safe	?
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)					
M2	`input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri` (local=identity); storage-fake tests			M1 safe	?
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280)					
M3	Configurable output/scratch base ? kill hardcoded `output/` ('trajectory_hybrid:237,313', `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`** ; golden + stub-E2E byte-identical			M1 ? default-OFF + Launch Report on flip	?
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282)					
M4	`DeviceContext` + WASB **CPU-fallback** ; unified healthcheck/device wiring			M1 safe (cuda default unchanged)	?
[#283](https://github.com/Khelsutra/badminton-highlight-indexer/pull/283)					
M5	**truly-local profile** end-to-end (preset + startup checks); first committed **runlog** (truly-local sample run)			M2,M3,M4 owner real-GPU	?
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-checks+L4 parity ?; runlog ? owner GPU)					
M6	Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +				

```

**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation +
`/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |
[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT
verify + owner_id tag, default-OFF ?; **exposure-safety boot posture** in `startup.py` ? edge-
auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see
changelog; consent cols + ToS/DPA ? owner/counsel) |
125 | M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit
owner approval | ? | ? |
126 | **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the
upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-
eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | **skeleton landed** (see changelog):
`backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook,
**consent faked-deny**; live activation ? owner/counsel (consent schema) + owned-model/golden
data |
127 | **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the
source from + writes the stitched reel **back to the user's Google Drive next to the source**
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? |
**backend landed** (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` +
`gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app ?
owner** |
128
129 **Testing strategy is a first-class deliverable** ?
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ?
[`runlogs/`](runlogs/).
130
131 ## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132 **? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:**
133 1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
134 3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
135 4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to
proxy).
136 5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data;
counsel-gated to M9).
137 6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until
the owned model clears the floor; capture?store?eval?goldenize meanwhile).
138
139 **? The remaining smaller picks are LOCKED (owner, 2026-06-21) ? see §2 D16:**
140 2. ? **Stitch placement ? co-located** in the detect job (= D4; one metered unit; the
job table supports a future CPU-split non-breaking, revisit only on bursty-compile metrics).
141 7. ? **RESOLVED ? D14:** cloud needs a **`gdrive-api` (OAuth) StorageBackend** (read the
source from + write the reel back to the user's Drive) ? the Northstar's Drive round-trip ?
distinct from the local **mount** backend (which stays for truly-local). New milestone **M12**.
142 8. ? **Codec ? D16:** **libx264 ENCODE** (protect parity) + **NVDEC DECODE-only**
(speed, no output-byte change); NVENC-encode deferred behind a config knob.
143 9. ? **Input cap ? D16:** **hard-reject`<2GB`** for v1, as a config cap (raise to 4GB
later in one line).
144 10. ? **Directory name ? `COMPUTE_DECOUPLED_SERVING/` CONFIRMED final** (already the
name everywhere + in ADR-014); `CLOUD_RUN_GPU_SERVING/` dropped.
145
146 ## 9. Definition of done

```

```

147 - [ ] M1?M4 land (config + input + output-base + device layer), each green on golden
cross-OS + experiment no-regression.
148 - [ ] **Profile-parity test green** ? truly-local vs cloud-mock produce byte-identical
windows + segment ranges.
149 - [ ] **truly-local** (M5) and **cloud-serving** (M7) each have a committed **runlog /
DEPLOY_REPORT** with a real sample run.
150 - [ ] Cost ledger (M8) makes GPU + **stitch** + egress cost visible per job; edge auth
(M9) gates multi-tenant.
151 - [ ] **Data-flywheel harness (M11)** captures?stores?shadow-evals?goldenizes consented
adult jobs; the owned model is on a measured path to the D11 floor.
152 - [ ] ? flip Status DONE, archive to `docs/archives/past_projects/`.
153
154 ## 10. References
155 **ADR lineage:** ADR-005/008/009/010/011/012/013 ? **ADR-014**
(`docs/archives/decisions/DECISIONS.md`). **Strategy:** `PLATFORM_ARCHITECTURE.md`,
`HOSTING_PLAN.md`, `VIDEO_LOCALITY_MODEL.md`, `DESKTOP_APP_PLAN.md`,
`LOCAL_APPLIANCE_ARCHITECTURE.md` (Khelsutra). **Archived predecessor:**
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (esp. `05-COST-ATTRIBUTION.md`, `07-COMPUTE-
ABSTRACTION.md`, `06-RISKS-AND-OPEN-QUESTIONS.md`, `DEPLOY_REPORT_GCP_2026-06-20.md`). **Code
anchors `[verified]`:** `backend/pipeline/detectors/base.py:164`, `tracknet_runner.py:281`,
`stitcher/compiler.py:303`, `segmenters/trajectory_hybrid.py:65,237`, `backend/storage/*`,
`backend/worker/__main__.py:88`, `backend/api/{uploads,job_worker}.py`,
`backend/infrastructure/database.py`. **Design provenance:** workflow `wf_56274ff0` (4 code-
mappers ? synthesis; 58 findings).
156
157 ---
158 ### Changelog
159 - 2026-06-21 ? created from the design workflow output; M0 in progress.
160 - 2026-06-21 ? **M0 APPROVED**, **M1 landed**
([#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279)):
`deployment.profile` + `compute_target` + preset bundles + precedence/auto-detect-as-suggestion
(D15) on `AppConfig`. Pure/additive, `profile=''` default == today's behaviour (zero core
change); presets touch only existing knobs (no M2 `input_backend` / M3 output-base over-reach).
Suite 1068 green, cov 84.11%.
161 - 2026-06-21 ? **M2 landed** ([#280](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/280)): `input_backend` / `input_root` + URI-aware main API/CLI input via
`resolve_input_uri` / `acquire_local_input` (local = identity passthrough, byte-for-byte
unchanged; bucket/Drive fetched to scratch). Unsupported scheme ? clean 400; `local://` accepted
on the resolved key; hosted-mode rejects non-local URIs; source URI kept for DB/report
provenance. Suite 1105 green, cov 84.84%.
162 - 2026-06-21 ? **M3 landed** ([#282](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/282)): configurable output/scratch base via
`backend/utis/paths.output_base(config)` ? the 3 segmenters' chunk/handoff/temp dirs root under
`output.output_dir` (default `output`, byte-identical; default-OFF ? no Launch Report). Fixes
the latent bug where scratch ignored `--output-dir`. Suite 1114 green, cov ~84.9%.
163 - 2026-06-21 ? **M4 landed** ([#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283)): `DeviceContext` + native WASB **CPU-fallback** via a new `device="auto"`
(cuda-if-available-else-cpu); explicit `cuda` stays strict (no silent slow-CPU downgrade),
default unchanged. `auto` resolves before argv (never reaches `wasb_infer.py`); CUDA probe
cached. Device-selection tested offline; real CPU-inference run owner/GPU-verified. Suite 1126
green, cov 84.96%. **? M1?M4 (the config/input/output/device layer) COMPLETE.**
164 - 2026-06-21 ? **M5+ batch authorized + design picks locked** (owner): the remaining $8
picks resolved ? **D16** (M6 = Cloud Run **Jobs** + bucket-poll; **libx264 encode + NVDEC
decode-only**, **hard-reject <2GB** input cap; dir name confirmed) and **D17** (real cloud
verification budget = **$100/?10k cap**, delete-after). Owner gave a blanket go to write+merge
the **default-OFF code** for every remaining gated milestone (M5,M6,M8,M9-auth,M9-consent-
cols,M11-plumbing,M12), one isolated PR each, reserving the gate for the real
run/spend/calibration/counsel. Grounded by gate-map workflow `wf_4849400f`.
165 - 2026-06-21 ? **REAL M7 cloud GPU run ?** (owner-authorized, within the D17 cap):
provisioned an **L4** (`g2-standard-4`, asia-southeast1-a; Mumbai stocked-out) ? native WASB on
the L4 ? **22 rallies** on the smoke clip ?
`gs://?/outputs/05e0e577/{intervals.csv,report.json}`; VM DELETED ($0, ~$0.21). ? the no-Gemini
cloud run needs `--set indexing.skip_ai_handoff=true` (the fully-local path emits WASB candidate
windows directly ? the long-flagged "0 rallies" cold-start fix). The reel **stitch** validated
locally (`VideoCompiler` ? 22-segment highlights, 85 MB). DEPLOY_REPORT runlog = the cloud-GPU

```

dry-run guide (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`).

166 - 2026-06-21 ? ****M9-auth code landed**** ([#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290)): `backend/api/auth.py` ? Cloudflare Access ****Cf-Access JWT verify**** (RS256: sig+exp+iss+aud against the team JWKS; lazy `PyJWT[crypto]`; injectable JWKS ? fully-offline tests) + `resolve\_tenant` ? tags `create\_job(owner\_id=?)`; `SecurityConfig.{edge\_auth\_enabled=False, cf\_team\_domain, cf\_access\_aud}` + CORS via `RALLY\_CORS\_ALLOW\_ORIGINS`. ****Default-OFF**** Security review (`wf\_9b4c71d0`) found NO bypass; locked alg=none + HS256-confusion rejection. ****? M9 ? ? the consent cols + ToS/DPA are owner/counsel; the consent migration is now v7 (M8 took v6).****

167 - 2026-06-21 ? ****M8 code landed**** ([#288](https://github.com/Khelsutra/badminton-highlight-indexer/pull/288)): per-job ****cost ledger**** ? `backend/billing.py` (pure `usage × pricebook` math, USD internal / INR display) + a ****v6 migration**** (NULLABLE cost columns on `jobs` + a versioned `pricebook` table seeded `placeholder-v0` at \$0; additive-NULl, idempotent) + `record\_job\_cost`/`get\_job\_cost`/`get\_video\_cost` + `GET /api/jobs/{id}/cost` + `/api/videos/{id}/cost` + `BillingConfig` (****default-OFF****) + a gated `\_maybe\_record\_job\_cost` hook. Adversarial review (`wf\_498b24af`) folded a ****major silent-GPU-under-bill**** (an unpriced `gpu\_type` is now surfaced, not invisible \$0) + the ****honest no-op**** caveat (usage emission is a follow-up ? cost 0 until wired) + breakdown reconciliation. Suite green, ruff+mypy clean. ****? M8 ? ? owner inserts real GCP prices (a new pricebook version) + the worker usage-emission is a follow-up; placeholder ships \$0 = no user-facing change. v6 is the shared bump (M9-consent cols extend it).****

168 - 2026-06-21 ? ****M6 code landed**** ([#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)): `backend/compute/` ComputeBackend registry ? `CloudRunCompute` dispatches a Cloud Run ****Job**** (injectable client) then ****bucket-polls**** a done-marker via `execution\_policy.poll\_until` + the storage layer (D16); `get\_compute\_backend` returns a backend ONLY for `compute\_target=cloud\_run` (else `None` ? the worker's in-process path, ****default-OFF****). The dispatched container runs ****inline**** (forced `local\_gpu`+native, never re-dispatch); the worker writes a `--done-uri` completion marker on success. `deploy/cloudrun/{Dockerfile,README}` = the L4 image (CUDA+ffmpeg+fonts+`wasb` conda env; weights staged per WASB-C3) + the M7 runbook. Adversarial review (`wf\_87b22fa4`) folded a ****blocker**** (cloud `fetch` needs a dst ? reads silently returned `{}` on gcs/s3) + a poll-wait test gap + `PolicyTimeout`?clean rc 4. Suite green, ruff+mypy clean. ****? M6 ? ? the real build/push/L4 run + DEPLOY_REPORT is M7 (owner, D17 budget).****

169 - 2026-06-21 ? ****M5 code landed**** ([#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286)): `backend/config/startup.py` ? per-profile fail/warn-fast startup checks (cloud-serving + non-cloud storage = the one fatal coherence error; GPU/creds mismatches = warnings, the runner healthcheck stays the authority), wired into the server (`\_enforce\_startup\_or\_exit` ? exit 1) + the worker (`cmd\_infer`/`cmd\_train` ? rc 2), default-OFF (probe gated behind an active profile ? the worker stays torch-free at `profile=""`). ****L4 profile-parity test**** (`tests/test\_profile\_parity.py`): same stub trajectory ? byte-identical windows + segment ranges across truly-local / cpu-mock / no-profile + worker-vs-in-process. Adversarial 4-lens review folded (`wf\_34671ac1`). Suite green, ruff+mypy clean. ****? M5 ? ? the real-GPU truly-local runlog (L5) is owner-side**** (template pre-staged in `runlogs`/).

170 - 2026-06-21 ? ****M9 hardening: exposure-safety boot posture**** (alpha-shareable "tunnel-to-a-box" enabler): extended `backend/config/startup.py` with a ****server-only exposure posture**** (new `include\_exposure=True` from `\_enforce\_startup\_or\_exit`; the worker stays unaffected ? it reads `--video-uri`, not untrusted tenant paths). Gated on `security.edge\_auth\_enabled` (****default-OFF ? strict no-op****, byte-identical to today): when the owner flips edge auth ON to expose the engine behind the CF-Access gate, three checks fire at ****boot**** ? `EXPOSED\_NO\_VIDEO\_ROOT` (****fatal****: `allowed\_video\_root` jail unset ? a tenant could read an arbitrary local file), `EXPOSED\_EDGE\_AUTH\_INCOMPLETE` (****fatal****: CF team-domain/AUD missing ? lifts the current first-request 401 to boot), `EXPOSED\_AUTH\_EXTRA\_MISSING` (****warn****: `PyJWT[crypto]` not importable ? would 500 per request). So a half-configured exposure fails fast instead of leaking/500-ing. +15 unit tests (default-OFF no-op, worker-ignores, both-fatals-compose-with-profile-checks, server-exit, WARN-only-no-exit). Adversarial 3-lens review (`wf\_35b735a9`) confirmed default-OFF byte-identical + no bypass; folded a ****whitespace-drift**** (aligned `auth.resolve\_tenant` to `.strip()` team/AUD so "configured" means the same at boot + at request time) + named the config fields in the edge-auth-incomplete message. Pairs with the appliance ****T3**** network posture (raw `:8000` refused ? owner-run). Suite green, ruff+mypy clean. ****? backstops M9 ? the actual exposure flip + CF team/AUD/ingress + key rotation stay owner-side. (Deferred to the PR-2 runbook: document that `upload\_dir` must sit under `allowed\_video\_root`, and that you MUST set `edge\_auth\_enabled=True` when exposing ? these checks only fire then).****

171 - 2026-06-22 ? ****M11 skeleton landed**** (data-flywheel harness, D12): `backend/flywheel.py` (`capture\_job` ? per-video `Videoworkspace` via `backend.storage.put`;

```

`shadow_eval` ? `experiment.compare`/`compare_unlabeled`; `propose_promotion` ?
`promotion.promote_if_served_safe` with Gemini=control/owned=proposal; `run_for_job`
orchestrator) + `FlywheelConfig` (**default-OFF**) + a gated `_maybe_run_flywheel` job-
completion hook mirroring M8's cost hook. **TWO independent guards keep it INERT in
production:** `flywheel.enabled=False` (default) AND `consent_attested` is a **FAKED hard-deny**
(returns False for every job) ? so even enabled it captures **nothing** until the real consent
schema lands. This is the PLUMBING only; it reuses the real workspace/eval/promotion/storage
code so it works the day consent + the owned model + golden labels are wired. **NO DB
migration** (the skeleton persists no consent). +13 tests (default-OFF no-op, enabled-but-
consent-denied no-op, consented-capture writes the manifest, delegation of shadow-eval/promote,
the hook is best-effort). Suite green, ruff+mypy clean. **? M11 ? ? LIVE activation is
owner/counsel (the consent ToS/DPA + schema ? the consent columns) + owner/GPU (the owned model
+ golden corpus for shadow-eval/promotion). ? the future M9-consent migration is now `v9` (v7 =
`locale`, v8 = the compute-picker `compute` column ? the schema ladder drifts as siblings land;
check `PRAGMA user_version` before authoring).**
172 - 2026-06-22 ? **M12 backend landed** (`gdrive-api://` OAuth/API StorageBackend, D14
Northstar): `backend/storage/gdrive_api.py` `GDriveApiStorage(StorageBackend)` ? the headless
**Drive-API** round-trip (read source + write reel back), distinct from the mount-only
`gdrive://` (left untouched). Path-keyed `gdrive-api://<path>` resolves each component ? a Drive
file id via `files().list`; implements `fetch_to_local` (`get_media().execute()` bytes), `put`
(folder-creating `files().create` + a `_media_body` seam), `exists`/`stat`/`list`/`delete`.
Wired the scheme into `ref.py` (`_KNOWN` + the path-keyed branch), `__init___.backend_class`,
and the `StorageConfig.backend`/`input_backend` Literals + a `gdrive_api` settings dict (read
via a **hyphen?underscore** lookup in `get_storage`). **All Google libs lazy-imported** (`import
backend.storage` never pulls them); a `service` is **injectable** ? fully-offline tests via an
in-memory fake Drive (round-trip + scheme + registry + the underscore-settings lookup);
`_build_service` (ADC over the Drive scope) raises a clear `storage-gdrive-api`-extra error when
libs/creds are absent. **NO DB migration.** Adversarial 2-lens review (`wf_ae8c9b0f`). Suite
green (1466), ruff+mypy clean. **? M12 ? ? the LIVE per-user interactive OAuth app (consumer
"connect your Drive": client id/secret + refresh tokens) is owner-gated; the backend ships the
service-account/ADC build + the fake. Reads no secrets from config (creds via ADC only).**
173

```

5. user (2026-07-03T16:45:01.188Z)

```

1      # Studio media lifecycle ? ingest ? store (BYO medium) ? annotate ? train
2
3      > **Status:** `IN PROGRESS` ? **storage half COMPLETE**, decisions locked . **Owner:**
Avi . **Created:** 2026?07?01
4      > **Current (2026?07?01):** ? DONE ? P0, E2E harness, P1 (vault free?space), **P2
(engine free?space guard)**,
5      > **P3 (capabilities?storage)**, **P4 (store?into?medium `POST /api/assets`)**, **P5
(SPA medium picker)**, P6 (library),
6      > P7 (vended annotator core), P8 (annotate?in?UI + engine P8a + D4 proxy), P9
(collector MD5 + `import_annotations`/maturity
7      > + write?safety); **5 screencasts** (00 baseline . 01 library . 02 current?video . 03
annotate . **04 store?in?my?medium**).
8      > The whole **"add a video ? store it in the user's chosen medium ? free?space checked ?
appears in my videos"** flow is now
9      > end?to?end. ? REMAINING = owner?gated only: **P10/P11** (corpus promotion +
train/re?eval, DEFERRED D13) + the **`L`**
10     > beta follow?ups. $7 is the live tracker.
11     > **Lifecycle:** `DRAFT` ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
12     > **Tracking anchors:** $7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md`.
13     > **Relation:** builds on `LOCAL_APPLIANCE_ARCHITECTURE.md` (ingest?as?source?menu, D7),
`PER_RALLY_SELECTION.md`
14     > (RallyPicker), and `ANNOTATE_AND_TRAIN.md` (CSV?no?transform, MD5 join, **no "Train
now" button** D4). Consumes the
15     > engine's `backend/storage` + `backend/annotations` + `backend.worker`; the vault's
`storagectl`; the collector's
16     > `import-local`/`export`; and the MIT `rally-annotator`. Cross?repo work items are
proposals to coordinate, not written unilaterally.
17     > **Honesty note:** engine/vault/collector claims tagged `[verified]` (read against

```

```

`origin/master` this session ? the local
18     > checkouts were stale and were fast?forwarded first), `[net?new]` (to build), or
`[design]`. Not legal advice.
19
20     ---
21
22     ## 0. TL;DR
23     Turn five independent CLIs into ONE seamless KhelSutra Studio flow: **Add a video ?
store it in a user?chosen medium
24     (bucket S3/GCS, a Google Drive folder, or locally) ? (free?space checked) ? index ?
select + annotate in the UI ?
25     contribute the labels ? (offline, owner?gated) grow the corpus + re?eval the
segmenter.** Most of the *storage/ingest*
26     plumbing already shipped on the engine `origin/master` (URI ingest, capabilities, a
video/reel library, a cost ledger,
27     `storage.put` to any medium, real Google Drive). The **net?new** work is: a **free?space
guard** (the vault + engine), a
28     Studio **storage?medium picker + "my videos" library UI**, an in?UI **annotation** step
(embed the MIT annotator +
29     persist labels), and the **annotate?corpus?train bridge** (collector MD5/maturity +
vault promotion). Near?term "improve
30     the model" honestly means **a growing golden corpus + an offline ?F1 readout** ? not a
one?click served model
31     (`gate_verdict` still refuses a commercial ship; real training needs GPU + native WASB).
32
33     ## 1. Foundation ? what already exists `[verified origin/master, 2026?07?01]`
34     The engine did the storage heavy?lifting since the first sketch of this plan; do **not**
rebuild it.
35
36     | Capability | Where | State |
37     |---|---|---|
38     | URI?driven ingest ? `POST /api/process` **resolves a path OR
`s3:///`/`gcs:///`/`gdrive:///` URI** | engine `main.py:970?1034` (`storage.resolve_input_ref`
:994; local?only `os.path.exists` :1005); worker fetches via `acquire_local_input`
(`storage/__init__.py:72`) | ? done |
39     | `GET /api/capabilities` . `/api/health` . `/api/version` | engine
`main.py:403,365,393` (reports
segmenters/ffmpeg/gpu/gemini/wasb/`compute_target`/`cloud_available`) | ? done ? **but does NOT
report storage backends** |
40     | Video **library** + reels list + cost ledger | engine `/api/videos` :447, `/api/reels`
:461/:494, `/api/jobs/{id}/cost` :1037, `/api/videos/{id}/cost` :1052 | ? done; SPA now consumes
`/api/videos` (P6); `/api/reels` in the SPA is a follow?up |
41     | **`storage.put` to any medium** + backends local/s3/gcs | engine
`storage/__init__.py:156`; `base.py`, `local.py`, `s3.py`, `gcs.py` | ? done (S3/GCS first?class
+ tested; **no secrets in config** ? ambient boto3/ADC) |
42     | **Google Drive** ? `gdrive:///` real (not a stub) | engine `storage/gdrive.py` (mounted
Drive?for?Desktop alias) + `storage/gdrive_api.py` (Drive API) | ? done (mount path needs
Drive?for?Desktop; headless ? bucket) |
43     | Annotation **provider registry** | engine
`backend/annotations/{base.py,automated_provider.py}` (`AnnotationProvider` ? `(start,end)`
intervals) | ? scaffolding (no HTTP surface / no human?in?UI provider) |
44     | Chunked upload; process?poll?query?compile?download; compute?picker (local vs cloud
GPU) | engine `backend/api/uploads.py`; SPA `IngestPanel`/`useUpload`/`useIngestJob`; SPA
#104/#108 | ? done |
45     | **Durable corpus custody** (add/verify/dedup/evict, N+1, `corpus_index.json`) | vault
`storagectl/storagectl.py` (`cmd_add:525`, `BACKENDS:365`) | ? done; `sync/stage/proxy` still
**stubs** (`:612`) |
46     | Rally annotator (the CSV producer) | `rally-annotator` ? VLC Lua + Chrome MV3; pure TS
`csv.ts`, `mountPanel()` `panel.ts:104` | ? MIT; CSV byte?compatible VLC/browser/collector |
47     | CSV ? curated DB ? indexer manifest | collector `curate.py` `import_local:97`,
`export_eval_set:548` (typed `EvalSetManifest`) | ? ingests the annotator CSV with no transform
|
48
49     ## 2. Decisions locked
50     | # | Decision | Rationale |

```

```

51 |---|---|---|
52 | D1 | **Two storage planes, not merged.** Engine `backend/storage` = per?request
byte?mover (`fetch`/`put`/`signed_url`); vault `storagectl` = durable custodian + **sole
writer** of `corpus_index.json`. They meet only at contribution (`collector export` ?
`storagectl add`). | Different jobs; a merge is a risky rewrite |
53 | D2 | Product "store in my medium" = engine `StorageBackend.put`. Vault stays
Khe!Sutra's **internal training?corpus** custodian. | The engine already `put`s to s3/gcs/gdrive
|
54 | D3 | **Identity = whole?file MD5** (`compute_video_id`) at every hop; add a cross?repo
`video_id == compute_video_id` guard test. | The join key; must never drift |
55 | D4 | **Annotate a frame?identical proxy, key labels to the master MD5.** | A 4K master
isn't browser?seekable; ADR?002 proxy keeps frame?exact timestamps |
56 | D5 | New control?plane endpoints on the **engine FastAPI** (APIRouter, `uploads.py`
precedent). MIT `rally-annotator` gets **only** generic injectable seams. | Keep Studio glue in
the private repos; MIT boundary clean |
57 | D6 | `POST /api/process` gains a **`storage_uri` field** (don't overload
`video_path`). | Cleaner for the typed SPA client |
58 | D7 | **BYO?bucket = self?host posture now**; hosted BYO deferred (needs per?user creds
? the no?secrets model uses the server's ambient creds). | `allowed_video_root` jail rejects
URIs on the exposed alpha |
59 | D8 | **Google Drive via the mounted?Drive alias now**; a browser OAuth folder?picker
is later. | `gdrive.py` works today with Drive?for?Desktop; no OAuth plumbing exists |
60 | D9 | **Adopt a label maturity ladder now** (`needs_review?confirmed?gold`); stop
hardcoding `status=CURATED`. UI labels never auto?promote to golden. | Safety for a
UI?contributed corpus |
61 | D10 | **Embed the MIT labeler in Studio** (vs. run the extension). | The most seamless
single?origin flow; MIT allows it |
62 | D11 | Free?space: **BLOCK** local + WSL frame?stage; **WARN** cloud (buckets aren't
capacity?bounded ? cost/quota warn). | The real hazard is local disk + WSL extraction |
63 | D12 | Near?term "improve the model" = **corpus growth + offline ?F1**; **no "Train
now", no served swap** (D4 of `ANNOTATE_AND_TRAIN`). | `gate_verdict` refuses ship; real
training = GPU + native WASB |
64 | D14 | **P2?P5 medium?track decisions RATIFIED (owner 2026?07?01), so the track is
turnkey:** (a) the BYO?bucket/Drive picker is **self?host posture only** (offered only when no
`allowed_video_root` jail); the hosted alpha stays upload?to?local until per?user creds
(confirms D7). (b) ***"Store in my medium" = copy a durable, MD5?keyed copy into the chosen
medium** (`storage.put`), keeping a local working copy for the run; a pasted bucket/`gdrive://`
URI is already at?rest. (c) **Google Drive via the mounted Drive?for?Desktop alias now**
(confirms D8; browser?OAuth picker stays the `L` follow?up). (d) Free?space: **block local + WSL
(~15x source headroom), warn cloud** (confirms D11). | Removes every owner gate on P2?P5 ? the
next session builds straight through |
65 | D13 | **Annotate?in?UI (P8) is APPROVED** (owner 2026?07?01) ? build the P7?P8 track
now. Ships behind a **prod feature flag** (default?OFF) until consented / no?minors footage is
in hand; demos run on the mock/synthetic clip. **Contribute?train (P10/P11) is DEFERRED** by the
owner. | Unblocks the marquee annotation flow while keeping real?footage exposure
owner?controlled; the training loop waits |
66
67 ## 3. Verified net?new gaps (the actual work)
68 > *Snapshot as of project start (2026?07?01); $7 is the live status. **All storage gaps
now CLOSED:** #1 vault?half (P1) +
69 > engine half (P2) . #2 capabilities?storage (P3) . #3 store?into?medium (P4) . #4
library (P6) + medium picker (P5) +
70 > annotation UI (P8) . #5 annotator embed (P7, vendored) . #6 collector md5 + maturity
(P9); the D4 proxy (P8) too. Nothing
71 > here remains open ? only owner?gated P10/P11 + the `L` beta follow?ups.*
72
73 1. **No general free?space guard.** `[verified]` vault `cmd_add` (`storagectl.py:525`)
copies with no `shutil.disk_usage`
74 check; the engine's only capacity check is a **WSL?only, warn?only, default?0**
frame?extraction guard
75 (`detectors/base.py:wsl_free_gb`; `wasb_runner.py:165`, `tracknet_runner.py:184`) ?
the real disk?filling peak, still
76 not blocking. No `upload_dir`/scratch guard, no HTTP 507. *The explicit owner ask.*
77 2. **`/api/capabilities` doesn't report storage backends** `[verified main.py:423?444]`
? the medium picker has no honest

```



```

78     source of truth for which media THIS box can use.
79     3. **No "store into my medium" flow** `[net?new]` ? `storage.put` exists but no endpoint
wires *upload ? put?to?user?medium
80     ? register as a library asset*; today an upload lands in `upload_dir` and processing
fetches/identity?passes.
81     4. **SPA has no medium picker, no "my videos" library, no annotation UI** `[verified]` ?
`IngestPanel` is one upload
82     button + one URI box; `client.ts` calls
`/process,/jobs,/query,/compile,/config,/capabilities,/upload/*` only (not
83     `/videos`,`reels`); `App.tsx` loads a single `?video=`.
84     5. **Annotator isn't embeddable / has no server sink** `[verified]` ? web labeler
persists to IndexedDB + downloads;
85     `PersistFn` is **synchronous** (a network POST can't report success ? silent?drop
risk); `deriveIdentity` yields only
86     `yt:`/`url:`; not a corpus id.
87     6. **Collector export omits `md5`** `[verified curate.py:639?651]` and **hardcodes
`status=CURATED`** (`:209`); no
88     `labeled_window`. The indexer join is a manual promise; UI labels would auto?promote
to golden.
89
90     ## 4. Design / architecture
91
92     ### 4.1 End?to?end flow
93     ```
94     Medium picker (local|s3|gcs|gdrive) ?? POST /api/assets {source_uri|uploaded_path,
medium_uri, sport}
95     local upload ? chunked /api/upload/* (free_bytes: upload_dir ? 507)          store:
StorageBackend.put ? medium
96     bucket/Drive ? resolve_input_ref
register: video_id=MD5 (library)
97     "My videos" ?? GET /api/videos ??
98     ?? Index ?? POST /api/process {storage_uri} (fetch?scratch; free_bytes: WSL stage
BLOCK) ?? jobs ? /api/query ? RallyGrid
99     ?? Annotate ?? GET /api/source/{id} (FileResponse of a frame-identical PROXY on the
local box)
100     embed rally-annotator mountPanel({video, identity=video_id,
persist=async POST})
101     Save ?? POST /api/annotations {video_id, rows[]} ? storage.put
labels/<id>.rallies.csv (no transform)
102     ? import_annotations(source=studio,
maturity=needs_review) [collector lib]
103     ? (owner-gated) collector export(+md5) ?
storagectl add ? gs://?rally-corpus (N+1)
104     backend.worker train / reeval ? weights/<ver>/
+ ?F1 (NO auto-ship)
105     ```
106
107     ### 4.2 Free?space guard (per medium)
108     One primitive `free_bytes()` on **both** backend contracts; one
`require_free()`/HTTP?507 caller.
109
110     | Medium | Measure | On fail |
111     |---|---|---|
112     | local (upload_dir, scratch, **WSL frame?stage**) | `shutil.disk_usage(target).free` vs
`need = f(stage)` (raw ? source; +proxy; extraction ? ~15× source) | **BLOCK** (507 / vault
refuse) |
113     | S3 / GCS | `None` ? verify write perm + **cost/quota warn** | **WARN** |
114     | Google Drive (mount) | free bytes of the mount fs; Drive API `about.storageQuota` when
the API backend is used | **BLOCK** on real quota |
115
116     ### 4.3 Annotation in Studio
117     Embed the MIT `csv.ts`/`annotator.ts`/`mountPanel` (allowed by MIT; preserve the
notice). Play a **frame?identical
118     proxy**, key the CSV to the **master MD5** (D4). On the single box serve via a server
**FileResponse** (`LocalStorage` has

```

```

119     no `signed_url` ? `base.py` raises); signed URLs only in hosted posture, and only with
bucket **CORS + HTTP Range** (else
120     seeking breaks) ? note a bearer signed URL **bypasses Cloudflare Access**. Make
`PersistFn` **async** (a generic,
121     MIT?justified change) so a failed save is loud. Studio constructs the `VideoIdentity`
from `video_id` (don't rely on
122     `deriveIdentity`). VLC stays the local?file power path; the Chrome extension stays for
arbitrary web video.
123
124     ### 4.4 Annotate ? corpus ? train (honest scope, D12)
125     `POST /api/annotations` ? collector `import_annotations` (pure?fn refactor of
`import_local`, CVAT?parity validators,
126     `status=needs_review`, persist source MD5 + `labeled_window`) ? owner?gated promotion
`collector export(+md5)` ?
127     `storagectl add --to F,G` (vault = sole `corpus_index.json` writer, N+1) ? `POST
/api/jobs {type:train|reeval}` shells
128     `backend.worker train` / `rally_seg_eval` and surfaces **?F1**. `gate_verdict` still
guards any ship; no "Train now".
129
130     ## 5. Threat model / privacy / licensing
131     - **MIT boundary:** `rally-annotator` receives only generic injectable seams (async
`PersistFn`, `mountPanel`, `csv`,
132     `i18n`); all Studio URLs/auth/asset?schema live in the private repos. Every new dep
stays MIT/Apache/BSD.
133     - **Consent / no?minors:** footage served to a browser `` stays behind the edge
gate; record a **consent/no?minors
134     provenance flag on the asset** (a data?model field, not a human promise). Avoid
signed?URL serving of raw footage on the
135     alpha (bypasses Access).
136     - **Paid?LLM?not?training:** UI (human) labels are compliant; a
machine?suggested?then?accepted label (the dormant
137     `candidate_segments.human_label` correction loop) must be tagged distinctly so the
`no_paid_llm_training_labels`
138     attestation holds.
139     - **BYO?bucket creds:** the no?secrets model uses the **server's** ambient creds ? "my
bucket" = "a bucket the server SA
140     can reach"; true per?user BYO needs scoped credentials (a beta item).
Self?host/single?tenant only for alpha (D7).
141
142     ## 6. Observability (first?class, owner directive)
143     Thread `video_id`/`upload_id`/`request_id` through every new path; structured logs at
entry/result/error; loud failures,
144     no silent drops (esp. the async annotation POST); a log?analysis pass per PR (per
`LOCAL_APPLIANCE_ARCHITECTURE.md §5`).
145
146     ## 7. Deliverables & progress tracker ? **source of truth**
147     Legend: ? Todo · ? In progress · ? Done · ? Owner?gated. **One small PR per row**,
branched from `origin/master`, no
148     stacking; CI green gates merge; **coverage ? floor and rising**; each PR ships its tests
+ meaningful logs.
149
150     | ID | Deliverable | Repo | Dep | Gated | Status | PR |
151     |----|-----|-----|-----|-----|-----|----|
152     | P0 | This tracked doc + `docs/README.md` index + `NEXT_STEPS.md` pointer | khelsutra |
? | no | ? | [#120](https://github.com/avidullu/khelsutra/pull/120) |
153     | E2E | **Screencast / e2e harness** ? single?origin mock engine (`web/e2e/mock-
engine.mjs`) + Playwright CUJ (`*.pw.mjs`) recording the tool on a local server ? `khelsutra-
screencast/`. The **checkpoint?as?screencast** mechanism; extended per feature. (`npm run
screencast`) | khelsutra | ? | no | ? | [#122](https://github.com/avidullu/khelsutra/pull/122) |
154     | P1 | **Free?space guard (vault):** `Backend.free_bytes(tier)` + pure
`capacity_status`; `cmd_add` blocks fs / warns cloud; per?tier `min_free_gb` margin (+10 tests).
*(partial?file cleanup on verify?fail ? follow?up)* | rally?corpus?vault | ? | no | ? |
[#2](https://github.com/Khelsutra/rally-corpus-vault/pull/2) |
155     | P2 | **Free?space guard (engine):** `require_free_bytes` helper (507) in `upload_init`
+ fetch scratch; **flipped `min_free_gb` WSL guard warn?block**

```

```
(`base.py::wsl_frame_stage_shortfall`, ~15x source). **D14: block local + WSL, warn cloud.** |
badminton?highlight?indexer | ? | coord | ? | [#407](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/407) |
156 | P3 | `/api/capabilities` reports **usable storage backends** (local always; s3/gcs if
SDK+config; gdrive if mounted) via `backend/storage/capabilities.py`. Shape = the SPA mock's
`storage.backends` block (`web/e2e/mock-engine.mjs`). No secrets. | badminton?highlight?indexer
| ? | coord | ? | [#408](https://github.com/Khelsutra/badminton-highlight-indexer/pull/408) |
157 | P4 | **Store?into?medium**: `POST /api/assets` + free?space precheck (P2) + register
(video_id=MD5). **D14: COPY a durable MD5?keyed copy into the medium via `storage.put`, keep a
local working copy; a pasted URI is already at?rest; SELF?HOST posture only (403 when
`allowed_video_root` set); edge?auth mirrors `/api/process`.** | badminton?highlight?indexer |
P2,P3 | coord | ? | [#409](https://github.com/Khelsutra/badminton-highlight-indexer/pull/409) |
158 | P5 | **SPA storage?medium picker** in `IngestPanel` (renders only
capabilities?reported usable media; local upload \ | S3/GCS URI \ | Drive mount) +
`createAsset`/storage types in `api/client.ts` + `ingest.medium.*`/`storage.*` in all 6 locales
(parity + `copy:locales` mirror) + e2e CUJ ? screencast `04?store?in?my?medium?cu.mp4`.
Coverage rose (funcs 85.9?87.1). **D14: self?host posture only; Drive = mounted alias.** |
khelsutra | P3,P4 | no | ? | [#137](https://github.com/avidullu/khelsutra/pull/137) |
159 | P6 | **SPA "my videos" library** ? `VideoLibrary` consumes `GET /api/videos`; reopen a
stored video (reload?survival) across 6 locales (+tests, coverage up); screencast `01`.
*(reels?list `/api/reels` in the SPA ? follow?up)* | khelsutra | ? | no | ? |
[#124](https://github.com/avidullu/khelsutra/pull/124) |
160 | P7 | **rally?annotator embeddable library** ? vite lib build
(`mountPanel`+`csv`+`i18n`, no chrome), **async `PersistFn`, injectable identity/video |
rally?annotator | ? | no | ? | ? |
161 | P8 | **Annotate?in?Studio**: SPA `AnnotatePanel` over the vendored core (mark
start/end · reason · shots · save/undo · loud save?state), flagged `?annotate=1` (D13);
screencast `03`. **P8a REAL engine endpoints DONE** ? `POST /api/annotations` writes the
verbatim CSV to `output/labels/<video_id>.rallies.csv` (MD5?keyed) + `GET /api/source/{id}`
(Range?capable) serves a **frame?preserving proxy** (D4, [engine
#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406) ? `scale=-2:720`,
timeline/duration?parity verified, warmed in background; labels stay keyed to the master MD5),
source fallback. | khelsutra + engine | P6,P7 | ? APPROVED (D13; prod flag) | ? | SPA
[#128](https://github.com/avidullu/khelsutra/pull/128) · engine
[#405](https://github.com/Khelsutra/badminton-highlight-indexer/pull/405) · proxy
[#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406) |
162 | P9 | **Collector library + fields (DONE)**: MD5 in export manifest ([col
#51](https://github.com/Khelsutra/sports-data-collector/pull/51) ? the deterministic join key) ·
callable `import_annotations` + **maturity gate** ([col
#52](https://github.com/Khelsutra/sports-data-collector/pull/52) ?
`status=STAGING`=needs?review, no auto?promote) · **write?safety**: vault atomic manifest write
([vault #3](https://github.com/Khelsutra/rally-corpus-vault/pull/3)) + collector SQLite
`busy_timeout` ([col #53](https://github.com/Khelsutra/sports-data-collector/pull/53)).
`labeled_window` DEFERRED ? premature: nothing captures a reviewed window yet (annotator
records rallies, not a span) & the eval that'd consume it is deferred (P10/P11).* |
sports?data?collector + vault | ? | no | ? | #51 · #52 · #53 · vault #3 |
163 | P10 | **Corpus promotion glue**: collector export ? `storagectl add`; cross?repo
`video_id=compute_video_id` guard test | khelsutra/deploy + vault | P9 | ? DEFERRED (D13) | ? |
? |
164 | P11 | **Train/re?eval job**: `POST /api/jobs {type:train|reeval}` ? `backend.worker` +
?F1 surface (no served swap) | badminton?highlight?indexer | P10 | ? DEFERRED (D13) | ? | ? |
165 | L | *(later/beta)* shared `parse_uri` module · `corpus_index.json`?derived eval
manifest · native Drive OAuth folder?picker · per?tenant RLS + per?user BYO creds · remote?GPU
bridge | multi | ? | ? | ? | ? |
166
167 ## 8. Open questions ? owner
168 > *All P2?P5 gates RESOLVED 2026?07?01 (see D14): Q2 free?space, Q3 BYO?bucket posture,
Q4 Drive. Q1/Q5 already resolved. None block the medium track.*
169 1. ~~Confirm the **frame?identical?proxy** choice (D4).~~ ? RESOLVED ? owner approved;
shipped as a background?warmed, timeline?parity?verified proxy ([engine
#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406)).
170 2. Free?space **headroom multiplier** for WSL frame extraction + block?vs?warn per
surface (D11 defaults: ~15x, block local/WSL). *(still open; lands with P2.)*
171 3. BYO?bucket on the **exposed alpha**: keep upload?only until per?user creds exist, or
enable URI ingest self?host?only? (D7 rec: self?host only.)* (decide with P4/P5.)*
```

```

172 4. Google Drive: mounted?Drive alias sufficient for alpha (D8), or prioritize a
browser OAuth folder?picker? *(mount is the current call; OAuth is the `L` follow?up.)*
173 5. Adopt the maturity ladder now vs. defer.~ ? RESOLVED ? adopted in P9 ([col
#52](https://github.com/Khelsutra/sports-data-collector/pull/52);
`status=STAGING`=needs?review).
174
175 9. Definition of done *(? = partially met; live status)*
176 - [x] A user picks a medium (local/S3/GCS/Drive?mount), the video is stored there
after a free?space check, and appears in "my videos". *(medium picker P5 + store?into?medium
P4 + capabilities P3 + "my videos" P6; screencast 04.)*
177 - [x] Vault `add` refuses an over?capacity fs tier / warns cloud; engine upload returns
507; the WSL frame?stage guard blocks. *(vault P1; engine 507 + WSL block P2.)*
178 - [x] A user selects a stored video and completes rally annotations in the UI; the
CSV lands (no transform) keyed by `video_id`; a failed save is loud. *(P8 + P8a.)*
179 - [ ] ? Contributed labels flow via the collector (`md5` in manifest; `needs_review`) ?
`storagectl add` (N+1) ? a `reeval` shows F1, no served swap / no "Train now". *(collector
? P9; promotion + reeval = P10/P11, owner?deferred.)*
180 - [ ] Cross?repo `video_id=compute_video_id` guard test green; consent/no?minors
provenance recorded; MIT boundary intact. *(MIT boundary ?; guard test + consent field = P10 /
follow?up.)*
181 - [x] Every PR: CI green, coverage ? floor and rising, structured logs on new paths,
one?PR?per?item. *(held throughout.)*
182 - [x] Every meaningful UI checkpoint is proven by a recorded screencast (`npm run
screencast`); 5 shipped (00 baseline · 01 library · 02 current?video · 03 annotate · 04
store?in?my?medium).
183
184 10. References `[verified origin/master 2026?07?01]`
185 Engine: backend/main.py` (process :970, capabilities :403, videos :447, reels :461,
health :365, cost :1037/:1052);
186 backend/storage/{__init__.py (resolve_input_ref:48, acquire_local_input:72, put:156),
base.py, local.py, s3.py, gcs.py, gdrive.py, gdrive_api.py, ref.py}`;
187 backend/api/uploads.py` ; backend/annotations/{base.py, automated_provider.py}`;
backend/pipeline/detectors/{base.py:_wsl_free_gb, wasb_runner.py:165, tracknet_runner.py:184}`;
backend/config/models.py:64,99 (min_free_gb)` .
188 Vault: storagectl/storagectl.py` (`cmd_add:525`, `Backend:254`, `BACKENDS:365`,
can_evict:152`); schema/corpus_index.schema.json` ; STORAGE_MAP.md` .
189 Annotator: web/src/state/{csv.ts,annotator.ts}` , web/src/ui/panel.ts:104` ,
web/src/persist/store.ts` , vlc/rally_annotator.lua` .
190 Collector: src/cli/curate.py` (`import_local:97`, `export_eval_set:548`,
status=CURATED:209`); src/core/batch_preflight.py` ; src/parsers/cvat/rally_cvat.py` .
191 SPA: web/src/App.tsx` , web/src/api/{client.ts,types.ts}` ,
web/src/components/{IngestPanel,SetupWizard}.tsx` .
192 Peers: LOCAL_APPLIANCE_ARCHITECTURE.md` , PER_RALLY_SELECTION.md` ,
ANNOTATE_AND_TRAIN.md` .
193
194 Changelog
195 - 2026?07?01 (storage track COMPLETE) ? Shipped the whole "store?in?my?medium"
track: P2 engine free?space guard
196 ([engine #407](https://github.com/Khelsutra/badminton-highlight-indexer/pull/407) ?
require_free_bytes`/507 in
197 upload_init` + /api/process` remote fetch?to?scratch; flipped the WSL frame?stage
guard warn?BLOCK at ~15x source) ·
198 P3 capabilities?storage ([engine #408](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/408) ? a
199 secret?free storage.backends` block: local always, s3/gcs on SDK+config, gdrive on a
resolvable mount) · P4
200 store?into?medium ([engine #409](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/409) ? POST /api/assets` :
201 MD5?keyed storage.put` for uploaded+remote, register?in?place for local, at?rest?URI
no?copy; self?host 403 + edge?auth +
202 free?space 507) · P5 SPA medium picker (IngestPanel` renders only usable media,
createAsset` client + storage types,
203 ingest.medium.*`/storage.*` across all 6 locales + site mirror, e2e CUJ + screencast
04?store?in?my?medium?cuj.mp4` ;
204 coverage rose funcs 85.9?87.1). The storage half of the original ask is done; only

```

owner?gated P10/P11 + `L` remain.

205 - 2026?07?01 (accuracy pass) ? Reconciled the doc with shipped state: fixed **P2** (was mislabeled ? "next" ? ? not?started; pivoted to the annotate track per D13), added a **Current status** header line, marked \$3 gaps as a start?snapshot with a closed/open ledger, ticked the met \$9 DoD items, and struck the resolved \$8 questions (proxy #406, maturity #52).

206 - 2026?07?01 (build?out) ? Shipped **P1** (vault free?space

[#2](https://github.com/Khelsutra/rally-corpus-vault/pull/2)) . **E2E/screenshot harness**

([#122](https://github.com/avidullu/khelsutra/pull/122)) + portable

([#123](https://github.com/avidullu/khelsutra/pull/123)) . **P6** library

([#124](https://github.com/avidullu/khelsutra/pull/124)) . current?video label

([#127](https://github.com/avidullu/khelsutra/pull/127)) . **P7** vendored annotator core

([#126](https://github.com/avidullu/khelsutra/pull/126)) . **P8** annotate?in?UI (SPA

[#128](https://github.com/avidullu/khelsutra/pull/128)) + engine P8a

[#405](https://github.com/Khelsutra/badminton-highlight-indexer/pull/405)) + **D4 proxy**

([#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406)) . **P9** collector bridge (MD5 [col #51](https://github.com/Khelsutra/sports-data-collector/pull/51), `import\_annotations`/maturity [col #52](https://github.com/Khelsutra/sports-data-collector/pull/52)) + write?safety (vault atomic [#3](https://github.com/Khelsutra/rally-corpus-vault/pull/3), collector `busy\_timeout` [col #53](https://github.com/Khelsutra/sports-data-collector/pull/53)). 4 screenshots (00?03). Remaining: P2?P5 medium?picker track (unblocked); P10/P11 deferred.

207 - 2026?07?01 ? Initial `IN PROGRESS`. Multi?agent design pass (8 readers / 2 architects / adversarial critic); **re?grounded

208 against `origin/master` after discovering the local checkouts were stale** ? the engine already shipped URI ingest,

209 capabilities, the video/reel library, the cost ledger, `storage.put`, and real Google Drive, so the scope narrowed to

210 free?space guards + SPA medium?picker/library/annotation + the collector/vault contribution bridge.

211

6. assistant (2026-07-03T16:45:03.627Z)

Now let me search for the specific rows mentioned in the claim to extract exact text.

7. user (2026-07-03T16:45:05.070Z)

126:[Omitted long matching line]

8. user (2026-07-03T16:45:05.740Z)

109:| A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |

9. user (2026-07-03T16:45:06.399Z)

9:> end?to?end. ? REMAINING = owner?gated only: **P10/P11** (corpus promotion + train/re?eval, DEFERRED D13) + the **`L`**

10. user (2026-07-03T16:45:06.623Z)

164:| P11 | **Train/re?eval job**: `POST /api/jobs {type:train|reeval}` ? `backend.worker` + ?F1 surface (no served swap) | badminton?highlight?indexer | P10 | ? DEFERRED (D13) | ? | ? |

11. assistant (2026-07-03T16:45:09.673Z)

Let me get the full M11 and P10 rows more clearly:

12. user (2026-07-03T16:45:09.798Z)

110 ## 7. Deliverables & progress tracker ? **source of truth**

111 Legend: ? Todo . ? In progress . ? Done . ? Gated. **One small PR per row**, off `origin/master`, no stacking. Default-OFF where it touches core callers.

```

112
113 | ID | Deliverable | Depends | Gated? | Status | PR |
114 |---|-----|-----|-----|-----|----|
115 | M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? |
116 | M1 | `deployment.profile` + `compute_target` config + preset bundles +
precedence/auto-detect; unit tests (no behaviour change, profile=' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117 | M2 | `input_backend`/`input_root`; wire main CLI/API input through
`StorageRef.parse_uri` (local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118 | M3 | Configurable output/scratch base ? kill hardcoded `output/`
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119 | M4 | `DeviceContext` + WASB **CPU-fallback** ; unified healthcheck/device wiring | M1 |
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283) |
120 | M5 | **truly-local profile** end-to-end (preset + startup checks); first committed
**runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
121 | M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122 | M7 | **cloud-serving profile** e2e on L4: upload<2GB` + gdrive/bucket ? detect+stitch
on Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) |
M6 | ? owner (spend) | ? | ? |
123 | M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds,
wall_seconds, bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation +
`/api/.../cost` | M7 | ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission
? owner/follow-up) |
124 | M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`;
DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? |
[#290](https://github.com/Khelsutra/badminton-highlight-indexer/pull/290) (auth: Cf-Access JWT
verify + owner_id tag, default-OFF ?; **exposure-safety boot posture** in `startup.py` ? edge-
auth-on requires the `allowed_video_root` jail + CF team/AUD or the server refuses to start, see
changelog; consent cols + ToS/DPA ? owner/counsel) |
125 | M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit
owner approval | ? | ? |
126 | **M11** | **Data-flywheel harness (D12)**: **on a consented adult job, **capture** the
upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-
eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the
golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the
owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent
gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | **skeleton landed** (see changelog):
`backend/flywheel.py` + `FlywheelConfig` (default-OFF) + gated `_maybe_run_flywheel` hook,
**consent faked-deny** ; live activation ? owner/counsel (consent schema) + owned-model/golden
data |
127 | **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar)**: **cloud reads the
source from + writes the stitched reel **back to the user's Google Drive next to the source**
(per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local
mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? |
**backend landed** (see changelog): `backend/storage/gdrive_api.py` `GDriveApiStorage` +
`gdrive-api://` scheme + ADC build; hermetic fake-Drive tests. **Live per-user OAuth app ?
owner** |
128
129 **Testing strategy is a first-class deliverable** ?
[`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ?
[`runlogs/`](runlogs/).

```

13. user (2026-07-03T16:45:10.182Z)

```

146    ## 7. Deliverables & progress tracker ? **source of truth**
147    Legend: ? Todo · ? In progress · ? Done · ? Owner?gated. **One small PR per row**,
branched from `origin/master`, no
148    stacking; CI green gates merge; **coverage ? floor and rising**; each PR ships its tests
+ meaningful logs.
149
150    | ID | Deliverable | Repo | Dep | Gated | Status | PR |
151    |----|-----|-----|-----|-----|-----|----|
152    | P0 | This tracked doc + `docs/README.md` index + `NEXT_STEPS.md` pointer | khelsutra |
? | no | ? | [#120](https://github.com/avidullu/khelsutra/pull/120) |
153    | E2E | **Screencast / e2e harness** ? single?origin mock engine (`web/e2e/mock-
engine.mjs`) + Playwright CUJ (*.pw.mjs`) recording the tool on a local server ? `khelsutra-
screencast/`. The **checkpoint?as?screencast** mechanism; extended per feature. (`npm run
screencast`) | khelsutra | ? | no | ? | [#122](https://github.com/avidullu/khelsutra/pull/122) |
154    | P1 | **Free?space guard (vault):** `Backend.free_bytes(tier)` + pure
`capacity_status`; `cmd_add` blocks fs / warns cloud; per?tier `min_free_gb` margin (+10 tests).
*(partial?file cleanup on verify?fail ? follow?up)* | rally?corpus?vault | ? | no | ? |
[#2](https://github.com/Khelsutra/rally-corpus-vault/pull/2) |
155    | P2 | **Free?space guard (engine):** `require_free_bytes` helper (507) in `upload_init`
+ fetch scratch; **flipped `min_free_gb` WSL guard warn?block**
(`base.py::wsl_frame_stage_shortfall`, ~15x source). **D14: block local + WSL, warn cloud.** |
badminton?highlight?indexer | ? | coord | ? | [#407](https://github.com/Khelsutra/badminton-
highlight-indexer/pull/407) |
156    | P3 | `/api/capabilities` reports **usable storage backends** (local always; s3/gcs if
SDK+config; gdrive if mounted) via `backend/storage/capabilities.py`. Shape = the SPA mock's
`storage.backends` block (`web/e2e/mock-engine.mjs`). No secrets. | badminton?highlight?indexer
| ? | coord | ? | [#408](https://github.com/Khelsutra/badminton-highlight-indexer/pull/408) |
157    | P4 | **Store?into?medium**: `POST /api/assets` + free?space precheck (P2) + register
(video_id=MD5). **D14: COPY a durable MD5?keyed copy into the medium via `storage.put`, keep a
local working copy; a pasted URI is already at?rest; SELF?HOST posture only (403 when
`allowed_video_root` set); edge?auth mirrors `/api/process`.** | badminton?highlight?indexer |
P2,P3 | coord | ? | [#409](https://github.com/Khelsutra/badminton-highlight-indexer/pull/409) |
158    | P5 | **SPA storage?medium picker** in `IngestPanel` (renders only
capabilities?reported usable media; local upload \ | S3/GCS URI \ | Drive mount) +
`createAsset`/storage types in `api/client.ts` + `ingest.medium.*`/`storage.*` in all 6 locales
(parity + `copy:locales` mirror) + e2e CUJ ? screencast **`04?store?in?my?medium?cu4.mp4`**.
Coverage rose (funcs 85.9?787.1). **D14: self?host posture only; Drive = mounted alias.** |
khelsutra | P3,P4 | no | ? | [#137](https://github.com/avidullu/khelsutra/pull/137) |
159    | P6 | **SPA "my videos" library** ? `VideoLibrary` consumes `GET /api/videos`; reopen a
stored video (reload?survival) across 6 locales (+tests, coverage up); screencast `01`.
*(reels?list `/api/reels` in the SPA ? follow?up)* | khelsutra | ? | no | ? |
[#124](https://github.com/avidullu/khelsutra/pull/124) |
160    | P7 | **rally?annotator embeddable library** ? vite lib build
(`mountPanel`+`csv`+`i18n`, no chrome), **async `PersistFn`, injectable identity/video |
rally?annotator | ? | no | ? | ? |
161    | P8 | **Annotate?in?Studio**: SPA `AnnotatePanel` over the vendored core (mark
start/end · reason · shots · save/undo · loud save?state), flagged `?annotate=1` (D13);
screencast `03`. **P8a REAL engine endpoints DONE** ? `POST /api/annotations` writes the
verbatim CSV to `output/labels/<video_id>.rallies.csv` (MD5?keyed) + `GET /api/source/{id}`
(Range?capable) serves a **frame?preserving proxy** (D4, [engine
#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406) ? `scale=-2:720`,
timeline/duration?parity verified, warmed in background; labels stay keyed to the master MD5),
source fallback. | khelsutra + engine | P6,P7 | ? APPROVED (D13; prod flag) | ? | SPA
[#128](https://github.com/avidullu/khelsutra/pull/128) · engine
[#405](https://github.com/Khelsutra/badminton-highlight-indexer/pull/405) · proxy
[#406](https://github.com/Khelsutra/badminton-highlight-indexer/pull/406) |
162    | P9 | **Collector library + fields (DONE):** MD5 in export manifest ([col
#51](https://github.com/Khelsutra/sports-data-collector/pull/51) ? the deterministic join key) ·
callable `import_annotations` + **maturity gate** ([col
#52](https://github.com/Khelsutra/sports-data-collector/pull/52) ?
`status=STAGING`=needs?review, no auto?promote) · **write?safety**: vault atomic manifest write
([vault #3](https://github.com/Khelsutra/rally-corpus-vault/pull/3)) + collector SQLite
`busy_timeout` ([col #53](https://github.com/Khelsutra/sports-data-collector/pull/53)).
**`labeled_window` DEFERRED ? premature: nothing captures a reviewed window yet (annotator
records rallies, not a span) & the eval that'd consume it is deferred (P10/P11).* |

```

```
sports?data?collector + vault | ? | no | ? | #51 · #52 · #53 · vault #3 |
163 | P10 | **Corpus promotion glue**: collector export ? `storagectl add`; cross?repo
`video_id==compute_video_id` guard test | khelsutra/deploy + vault | P9 | ? DEFERRED (D13) | ? |
? |
164 | P11 | **Train/re?eval job**: `POST /api/jobs {type:train|reeval}` ? `backend.worker` +
?F1 surface (no served swap) | badminton?highlight?indexer | P10 | ? DEFERRED (D13) | ? | ? |
165 | L | *(later/beta)* shared `parse_uri` module · `corpus_index.json`?derived eval
manifest · native Drive OAuth folder?picker · per?tenant RLS + per?user BYO creds · remote?GPU
bridge | multi | ? | ? | ? | ? |
```

14. user (2026-07-03T16:45:10.442Z)

```
27 ## 2. Decisions locked
28
29 | # | Decision | Source / date | Implication |
30 |---|---|---|---|
31 | D1 | **Product alpha and owned-model launch are separate decisions.** |
`NEXT_STEPS.md` NO-GO banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled
hosted product loop without claiming the owned model is commercially cleared. |
32 | D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
```

15. assistant (2026-07-03T16:45:16.296Z)

Now let me check the exact context in §2, §4.6 of COMPUTE_DECOUPLED_SERVING and section 9 Q1 of ALPHA_LAUNCH_READINESS:

16. user (2026-07-03T16:45:16.460Z)

```
24 ## 2. Decisions locked
25
26 | # | Decision | Source | Implication |
27 |---|-----|-----|-----|
28 | D1 | The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile
branch inside it**. | design `[verified]` | All profile difference lives in config + the two
registries. |
29 | D2 | **Two registries:** `ComputeBackend` (where) selected by `deployment.profile` +
`compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`. | ADR-014
| Realizes PLATFORM_ARCHITECTURE's `local`|`hosted`|`byo_worker`. |
30 | D3 | **Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI);
`byo_worker` reserved/deferred. | ADR-014 | Enum/seam reserved so BYO isn't precluded. |
31 | D4 | In **cloud-serving, STITCH runs on Cloud Run** (co-located with detect) so
CPU/wall + egress are metered into the per-job cost ledger. | owner (a) | Stitch-on-laptop would
hide real cost. |
32 | D5 | **Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory);
**detect** is byte-exact only same-GPU. | deploy report 2026-06-20 | Cross-GPU asserted at the
rally/window level, never CSV bytes. |
33 | D6 | **Default-OFF, smallest-safe-first, one PR per milestone**; any core-caller
change (e.g. configurable output base) ships behind a flag + a Launch Report. | [[launch-report-
convention]] | Zero-regression discipline. |
34 | D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an
async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that
~$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane
only on a UX complaint. |
35 | D8 | **Pricebook = a versioned DB table**; cost computed in **USD** (matches GCP
billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore
market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |
36 | D9 | **Edge auth = Cloudflare Access** (reuse the site's existing CF). Lock the Cloud
Run **ingress to the CF proxy** + **verify the `Cf-Access` JWT signature** (so a direct hit to
the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity
system. |
37 | D10 | **Free-tier "data-for-reels" trade:** free reels in exchange for **training
rights** (uploaded footage + derived trajectories/labels); **paid tier = no-train** (privacy is
```


the paid feature). ****Carve-outs:**** train ONLY on attested ****ADULT**** footage (minor footage ? reel yes, training pool ****NO**** ? DPDP/GDPR need verifiable parental consent); train on ****DERIVED**** data + ****auto-delete raw****; ****ToS counsel-reviewed****; live training-on-uploads ****gated to M9**** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel. |

38 | D11 | ****Quality floor:** Gemini handoff ON** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a ****Launch Report****; the owned model runs in ****shadow/eval**** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |

39 | D12 | ****Data-flywheel harness is IN SCOPE (owner add, Q5):**** capture the (consented, adult) uploaded video + the Gemini reel/rally output ? ****reliably store**** in the per-video workspace/bucket ? ****run shadow evals**** (owned-vs-Gemini, dual-eval-set) ? ****promote good ones to the golden TRAINING set**** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new ****M11****; ties D10 (consent) + `data\_pipeline/` + the eval/experiment harness. |

40 | D13 | ****Profile is user-switchable ANYTIME ? a headline FEATURE.**** The same user runs ****local today, cloud tomorrow, back to local**** ? it's just a config/startup choice; the pure core gives ****equivalent**** output either way. ****Advertise it**** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the ****rally/window**** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is **between** jobs. |

41 | D14 | ****NORTHSTAR ? operate toward this:**** a ****mobile, app-like**** flow ? point at a ****Google Drive**** video ? ****fully cloud-native**** run ? the stitched, ****ready-to-share**** reel lands ****back in Drive, next to the source****. | owner 2026-06-21 | Implies a ****`gdrive-api` (OAuth) StorageBackend**** (read source + write reel back) ? distinct from the local ****mount**** backend; ****resolves \$8 #7****, new ****M12****. The mobile UI is the khelsutra track; the engine must support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive write); not necessarily v1, but the design must not preclude it (the ****StorageBackend**** ABC accommodates it). |

42 | D15 | ****Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.**** A GPU owner may still choose cloud; ****cloud (= spend) is never entered silently****; the active profile is always explicit + surfaced. | owner 2026-06-21 | Amends \$4.3 (auto-detect = a proposed default, not an auto-decision). |

43 | D16 | ****M6/M7 implementation SHAPE + the cheap \$8 picks (owner 2026-06-21):**** (a) ****Cloud Run JOBS**** (not Services-with-GPU) for the detect+stitch unit, ****bucket-poll for completion**** (reuses ****`JobWaiting`/`claim\_due\_job`** verbatim, fits async upload?poll?reel + scale-to-zero D7); (b) ****stitch ENCODE = libx264**** (protects the D5 byte-determinism/parity guarantee + the L4 parity test) ? ****NVDEC for DECODE only**** (a speed lever that never touches output bytes); ****NVENC encode is DEFERRED behind a config knob****, revisit only if M7 cost data justifies relaxing parity (\$8 Q8); (c) ****input cap = hard-reject `<2GB`**** for v1 (a config cap so raising to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? \$8 Q9). | owner 2026-06-21 | M6/M7. Resolves \$8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |

17. user (2026-07-03T16:45:16.527Z)

96 ### 4.6 The data-flywheel harness (D12 / M11) `[design]`

97 Gemini-on (D11) ships good reels now; this harness is ****how we earn the right to flip Gemini off****. On a ****consented, adult**** cloud-serving job (the D10 trade), the pipeline additionally:

98 1. ****Captures**** the source upload + the Gemini reel + the rally output (intervals/report) into the per-video workspace (****`workspaces/<video\_id>/`**), instead of the no-consent auto-delete path.

99 2. ****Reliably stores**** them durably in the bucket (the consented-retention bucket; raw video kept only for the training pool, not the no-consent path).

100 3. ****Runs shadow evals**** ? the owned model scores the same video alongside Gemini (****`backend/eval/experiment.compare`** + the dual-eval-set, [[eval-dual-set-regression]]) ? tracks the owned-vs-Gemini gap per job (the live read on "is the floor reachable yet?").

101 4. ****Promotes to golden**** ? captured videos that pass review become ****golden TRAINING rows**** (human-reviewed labels via the annotation flow ? ****`data\_pipeline/`**), growing the corpus ? the owned model climbs to the D11 floor ? flip Gemini off via a Launch Report.

102 This is the consent?capture?eval?goldenize ****flywheel**** (the cloud realization of the PERSONALIZATION contribution model): every consented free reel makes the owned model better. Reuses ****`data\_pipeline/`**, ****`backend/eval/`**, and the M9 consent gate; ****adults-only****, raw-retention gated by the D10 consent.

103

18. user (2026-07-03T16:45:16.597Z)

```
120
121     ## 9. Open questions
122
123     1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124     2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125     3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
```

19. assistant (2026-07-03T16:45:29.262Z)

Perfect. Now I have all the evidence I need. Let me analyze the key distinction:

****CONFIRMED FINDING:****

The claim is correct?there is a double-tracking of the same core engine-side flywheel work with ****no cross-reference**** and ****real drift risk****.

****Evidence:****

****M11 in COMPUTE_DECOUPLED_SERVING/README.md (\$7, row 126):****
> "Data-flywheel harness (D12): on a consented adult job, ****capture**** the upload + Gemini reel/rally output ? ****reliably store**** under the per-video workspace ? ****shadow-eval**** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? ****promote**** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow)... skeleton landed: `backend/flywheel.py`..."

****A12 in ALPHA_LAUNCH_READINESS.md (\$7, row 109):****
> "Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train|reeval` job, and delta-F1 surface"

****Supporting context from STUDIO_MEDIA_LIFECYCLE.md (\$7, rows 163?164):****
- ****P10****: "Corpus promotion glue: collector export ? `storagectl add`; cross-repo `video\_id==compute\_video\_id` guard test | ... DEFERRED (D13)"
- ****P11****: "Train/re-eval job: `POST /api/jobs {type:train|reeval}` ? `backend.worker` + ?F1 surface (no served swap) | ... DEFERRED (D13)"

****The critical distinction:**** M11 is the ****engine-side harness**** (capture ? store ? shadow-eval ? promote internally), while P10/P11 are the ****khelsutra Studio UI surfaces**** (the `api/segments` PATCH endpoint and the human-annotated corpus promotion to the vault). ****Both route through the same core flow**** (capture/eval/promote to golden), so a change in M11 (e.g., consent schema, eval methodology, promotion logic) will affect P10/P11 without cross-reference?exactly the "will drift" risk.

****The "decision row vs. implementation row" defence fails**** because A12 ***also*** lists deliverables (collector promotion, `train|reeval` job endpoints), not merely a decision to defer. It's a full tracking row with technical scope, making the overlap genuine.

****VERDICT: CONFIRMED****

Same deliverable double-tracked, no cross-link. M11 (\$4.6, `backend/flywheel.py` skeleton landed, default-OFF) is the capture?eval?promote engine harness; A12 proposes the Studio UI surface (P10/P11 corpus promotion + `train|reeval` endpoints + delta-F1 display). Both converge on "promote to golden TRAINING set" via the annotation flow, so M11 changes (consent, eval schema, promotion criteria) will silently affect A12 without acknowledgment in either tracker.

****Suggested fix:**** Add a one-line cross-reference in ****A12 row****: append "| See also: M11 (`backend/flywheel.py` harness) |" or reword the A12 description to "Reopen or defer P10/P11 ****UI surfaces**** for corpus flywheel (M11 engine-side capture/promote harness tracks the core)..."